Physica A 700 (2026) 131892

Contents lists available at ScienceDirect

Physica A

journal homepage: www.elsevier.com/locate/physa

Self-similar shrimp structures and bifurcation hierarchies in the Gumowski–Mira map

Daniel Borin [a,b,*], Diego F.M. Oliveira [a]

[a] University of North Dakota, School of Electrical Engineering and Computer Science, Grand Forks, ND 58202, USA
[b] São Paulo State University (UNESP), IGCE - Physics Department, 13506-900, Rio Claro, SP, Brazil

ARTICLE INFO

Dataset link: https://gumowskimira.streamlit.app

ABSTRACT

Mainly related to the aesthetic complexity of its attractors, the Gumowski–Mira map has attracted long-standing interest in nonlinear dynamics. Although the Gumowski–Mira map has been extensively studied in terms of its attractor properties, its global organization in the two-parameter space, including the arrangement and hierarchical organization of different stability structures associated with periodic and chaotic dynamics, has not been systematically investigated. In this work, we investigate the two-parameter space of the Gumowski–Mira map by combining Lyapunov exponent calculations with a systematic identification of the attractor period. We identify shrimp-shaped stability regions composed of a main periodic core followed by cascades of period-doubling bifurcations leading to chaos. Stability regions with irregularly shaped geometries but identical internal bifurcation organization are also observed. In addition, tongue-like periodic domains organized along paths of increasing period are detected. Both monotonic period-adding sequences and sequences involving skipped tongues are found, indicating that different mechanisms of period organization coexist.

1. Introduction

Nonlinear dynamical systems are extensively used to model countless phenomena across diverse scientific and technological domains, exhibiting a rich spectrum of dynamic states that range from steady states and regular (periodic or quasi-periodic) oscillations to chaotic fluctuations [1]. To obtain a comprehensive perspective of the long-range dynamics of systems and transitional structures and cascades that go beyond the scope of traditional single-parameter analysis, the exploration of two-parameter spaces is essential. This approach, which involves the simultaneous variation of two control parameters, allows for the use of tools such as isoperiodic diagrams and Lyapunov exponents to monitor and discriminate dynamic behaviors.

In the context of two-parameter space analysis, regions of regularity frequently emerge as islands within chaotic regimes. Among these structures, the term "shrimp" was coined by J. A. C. Gallas in 1993, while investigating the parameter space of the Hénon map [2], to describe stability islands with a characteristic shape, consisting of a central head and four narrow antennae [3–7]. The shrimp typically arises from the intersection of two parabolic arcs of superposition and has two distinct bifurcation lines surrounding the arcs [8]. Although it was only coined as a shrimp in 1993, these structures began to appear in the early 1980s [9–11] and were given various names, such as the fishbook [12], swallow [13], crossroads [14], and compound window [15].

The study of these structures is crucial for understanding the complex behavior of systems, as they represent regions of regularity (periodic or quasi-periodic oscillations) embedded within a chaotic sea. While periodic shrimps are the most prevalent and

* Corresponding author.
E-mail address: d.borin@hotmail.com (D. Borin).
URL: https://danielborin.github.io/ (D. Borin).

https://doi.org/10.1016/j.physa.2026.131892
Received 28 May 2026; Received in revised form 9 July 2026
Available online 11 August 2026
0378-4371/

extensively studied, recent research has uncovered the existence of quasi-periodic shrimps (associated with quasi-periodic oscillations or tori), which are relatively rare and typically observed in higher-dimensional phase spaces (at least three for maps) [16,17]. Furthermore, organized structures of periodicity, such as Arnold tongues [18], which emanate from rational frequency values and exhibit period adding sequences, are frequently observed in the parameter space of dissipative systems [19,20]. Recently, the study of system dynamics at the parameter space has attracted a great interest, including economics [21], ecology [22,23], cancer modeling [24], chemical reaction [25], laser [26,27], ecoepidemiology [28], population dynamics [29], fluid convection [30,31], neural network [32,33], geomagnetic field reversals [34], dynamics of gas bubbles in liquid [35]. Experimental studies were also carried out to clarify the real world existence of shrimps [36–39].

Beyond their fundamental importance for understanding nonlinear dynamics, the organization of parameter spaces and the characterization of periodic and chaotic attractors have important implications in a wide range of applications. For example, parameter-space analysis is closely related to multistability phenomena observed in nonlinear oscillators, where multiple attractors may coexist for the same set of control parameters [40]. Likewise, the accurate identification of dynamical regimes plays an important role in the synchronization of chaotic systems, including recent neural-network-based synchronization methods capable of operating under noisy and time-varying conditions [41,42]. These studies further illustrate the broad relevance of investigating parameter-space organization in nonlinear dynamical systems.

The Gumowski–Mira map has been investigated for several decades and has become a benchmark nonlinear system due to its rich variety of dynamical behaviors [43]. Beyond its fundamental importance, it has also found applications in chaos-based image encryption and secure communication systems, where its sensitivity to initial conditions and complex attractor geometries are exploited to generate cryptographically secure sequences [44,45]. Previous studies have addressed its formulation and general dynamical properties [46], attractor geometries and the characterization of different dynamical regimes [47], as well as complexity measures and indicators of chaotic behavior [48]. Nevertheless, the global organization of its two-dimensional parameter space, including the hierarchy of stability regions and the organization of shrimp-shaped structures and tongue periodic domains, has not been systematically investigated. Motivated by this gap, the present work investigates the parameter space of the Gumowski–Mira map by constructing Lyapunov exponent and isoperiodic diagrams. Our objective is to characterize the global organization of periodic and chaotic regions, identify self-similar shrimp structures and bifurcation hierarchies, and investigate the different parameter-space organizations observed in this system.

The paper is organized as follows. In Section 2, we introduce the Gumowski–Mira map and present its mathematical formulation, including the Jacobian matrix used for stability and Lyapunov analyses. Section 3 is devoted to the investigation of the parameter space, where bifurcation diagrams, Lyapunov exponent maps, and period diagrams are employed to characterize the global organization of dynamical regimes. Particular attention is given to the emergence of shrimp-shaped structures and period-doubling cascades. Finally, in Section 4, we summarize the main results and discuss their implications for the understanding of parameter space organization in nonlinear dissipative systems.

2. Model

The Gumowski–Mira map is known for creating beautiful attractors with patterns that look like natural objects [43]. It is a nonlinear system that shows a variety of different behaviors and has applied to model the longitudinal motion of particles in an accelerator. The system is described by a two-dimensional discrete map with a nonlinear damping term, and it is defined by the following set of equations [46,49,50]

$$\begin{cases} x_{n+1} &= a y_n \left(1 - b y_n^2\right) + y_n + F(x_n, \mu) \\ y_{n+1} &= -x_n + F(x_{n+1}, \mu) \end{cases} \tag{1}$$

where a, b, and μ are parameters that influence the system's dynamics, and the $F(x, \mu)$ is a nonlinear function called "bounded nonlinearities" and is defined as:

$$F(x, \mu) = \mu x + \frac{2x^2(1-\mu)}{1+x^2} \tag{2}$$

The nonlinearity in Eq. (2) ensures bounded growth for large $|x|$.

Fig. 1 shows representative attractors of the Gumowski–Mira map for several parameter values, confirming the visual richness commonly associated with this system. To explore additional parameter values, the supplementary material includes an interactive visualization of the attractors (see Section "Data Availability").

2.1. Fixed points

The equilibrium solutions of Eq. (1) are obtained by imposing

$$x_{n+1} = x_n = x^*, \qquad y_{n+1} = y_n = y^*. \tag{3}$$

Substituting these conditions into Eq. (1) yields

$$x^* = a y^* \left(1 - b y^{*2}\right) + y^* + F(x^*, \mu), \tag{4}$$

$$y^* = -x^* + F(x^*, \mu). \tag{5}$$

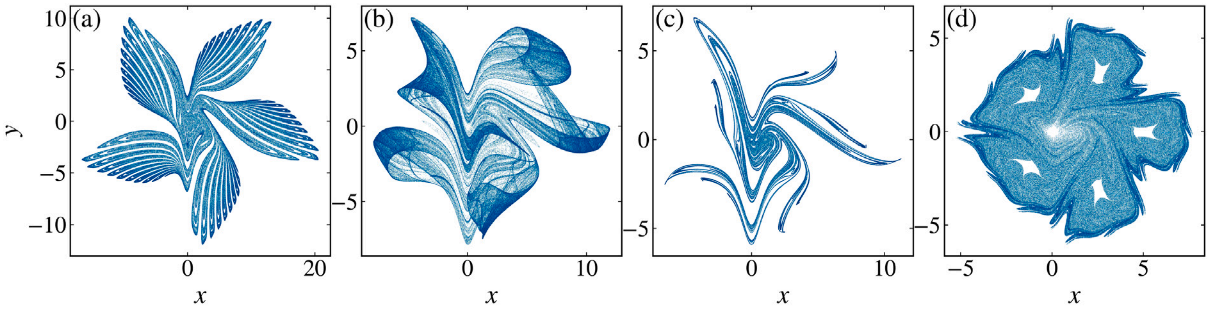

Fig. 1. Representative attractors of the Gumowski–Mira map for (a) $a = 0.01$, $b = 0.05$, $\mu = -0.8$; (b) $a = 0.85$, $b = 0.05$, $\mu = -0.8$; (c) $a = 0.01$, $b = 0.8$, $\mu = -0.8$; (d) $a = 0.01$, $b = 0.05$, $\mu = 0.21$; illustrating the diversity of dynamical behaviors and the visual complexity commonly associated with this system.

Eq. (5) can be rewritten as

$$F(x^*, \mu) = x^* + y^*. \tag{6}$$

Substituting Eq. (6) into Eq. (4) eliminates the nonlinear function and leads to

$$0 = ay^* \left(1 - by^{*2}\right) + 2y^*, \tag{7}$$

or equivalently,

$$y^* \left[a\left(1 - by^{*2}\right) + 2\right] = 0. \tag{8}$$

Therefore, the equilibrium solutions are divided into two branches.

The first branch is obtained from

$$y^* = 0. \tag{9}$$

In this case, Eq. (6) becomes

$$F(x^*, \mu) = x^*. \tag{10}$$

Using the definition of the nonlinear function,

$$F(x, \mu) = \mu x + \frac{2(1-\mu)x^2}{1+x^2}, \tag{11}$$

Eq. (10) can be written as

$$(\mu - 1)x^* + \frac{2(1-\mu)x^{*2}}{1+x^{*2}} = 0. \tag{12}$$

Multiplying by $(1 + x^{*2})$ gives

$$(1-\mu)x^* \left(x^{*2} - 2x^* + 1\right) = 0, \tag{13}$$

which simplifies to

$$(1-\mu)x^*(x^* - 1)^2 = 0. \tag{14}$$

Hence, for $\mu \neq 1$, this branch contains the two equilibrium points

$$(x^*, y^*) = (0, 0), \qquad (x^*, y^*) = (1, 0). \tag{15}$$

For the particular case $\mu = 1$, the nonlinear function reduces to $F(x, 1) = x$, and Eq. (10) is satisfied for any value of x.

The second branch is obtained from

$$a\left(1 - by^{*2}\right) + 2 = 0, \tag{16}$$

which yields

$$y^{*2} = \frac{1 + \frac{2}{a}}{b}, \tag{17}$$

provided that

$$\frac{1 + \frac{2}{a}}{b} > 0. \tag{18}$$

For these solutions, the corresponding value of x^* is determined from Eq. (6),

$$\mu x^* + \frac{2(1-\mu)x^{*2}}{1+x^{*2}} = x^* + y^*, \tag{19}$$

which is equivalent to the cubic equation

$$(1-\mu)x^{*3} + (y^* - 2 + 2\mu)x^{*2} + (1-\mu+y^*)x^* + y^* = 0. \tag{20}$$

Eq. (20) generally possesses one or three real roots, depending on the parameter values. Consequently, the second branch may generate additional equilibrium points whose existence depends on the parameters a, b, and μ.

2.2. Linear stability

The local stability of an equilibrium point is determined by the Jacobian matrix of the map evaluated at the fixed point. The Jacobian of Eq. (1) is defined as

$$J = \begin{pmatrix} \frac{\partial x_{n+1}}{\partial x_n} & \frac{\partial x_{n+1}}{\partial y_n} \\ \frac{\partial y_{n+1}}{\partial x_n} & \frac{\partial y_{n+1}}{\partial y_n} \end{pmatrix}, \tag{21}$$

whose elements are given by

$$\frac{\partial x_{n+1}}{\partial x_n} = F'(x_n, \mu), \tag{22}$$

$$\frac{\partial x_{n+1}}{\partial y_n} = 1 + a\left(1 - 3by_n^2\right), \tag{23}$$

$$\frac{\partial y_{n+1}}{\partial x_n} = -1 + F'(x_{n+1}, \mu)F'(x_n, \mu), \tag{24}$$

$$\frac{\partial y_{n+1}}{\partial y_n} = F'(x_{n+1}, \mu)\left[1 + a\left(1 - 3by_n^2\right)\right], \tag{25}$$

where

$$F'(x, \mu) = \mu + \frac{4(1-\mu)x}{(1+x^2)^2}. \tag{26}$$

Evaluating the Jacobian at an equilibrium point (x^*, y^*), for which $x_{n+1} = x_n = x^*$, yields

$$J^* = \begin{pmatrix} A & B \\ A^2 - 1 & AB \end{pmatrix}, \tag{27}$$

where, for convenience, we define

$$A = F'(x^*, \mu), \tag{28}$$

$$B = 1 + a\left(1 - 3by^{*2}\right). \tag{29}$$

The characteristic polynomial of the Jacobian is

$$\lambda^2 - T\lambda + \Delta = 0, \tag{30}$$

where

$$T = \mathrm{tr}(J^*) \tag{31}$$

and

$$\Delta = \det(J^*) \tag{32}$$

denote the trace and determinant of the Jacobian, respectively.

Using Eq. (27), the trace becomes

$$T = A(1+B), \tag{33}$$

whereas the determinant simplifies to

$$\Delta = A(AB) - B(A^2 - 1) = B. \tag{34}$$

Therefore,

$$\Delta = 1 + a\left(1 - 3by^{*2}\right). \tag{35}$$

An important consequence of Eq. (35) is that the determinant is independent of the nonlinear function $F(x,\mu)$. Consequently, the bounded nonlinearity influences the local stability only through the trace of the Jacobian, whereas the determinant depends exclusively on the parameters a, b, and on the equilibrium coordinate y^*.

The eigenvalues governing the local dynamics are obtained from Eq. (30) as

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4\Delta}}{2}. \tag{36}$$

The equilibrium is locally asymptotically stable whenever both eigenvalues satisfy

$$|\lambda_1| < 1, \qquad |\lambda_2| < 1. \tag{37}$$

For two-dimensional discrete maps, changes in the local stability occur when one or both eigenvalues cross the unit circle in the complex plane. These crossings define the local bifurcations of the fixed point and can be obtained directly from the characteristic polynomial without explicitly computing the eigenvalues.

2.2.1. Saddle–node bifurcation

A saddle–node (fold) bifurcation occurs when one eigenvalue crosses the unit circle through

$$\lambda = 1. \tag{38}$$

Substituting $\lambda = 1$ into Eq. (30) gives

$$1 - T + \Delta = 0. \tag{39}$$

Using Eqs. (33) and (35),

$$1 - A(1 + B) + B = 0, \tag{40}$$

which can be factorized as

$$(1 + B)(1 - A) = 0. \tag{41}$$

Therefore, provided that $1 + B \neq 0$, the saddle–node bifurcation is determined by

$$F'(x^*, \mu) = 1. \tag{42}$$

At this bifurcation two fixed points coalesce and disappear, producing the creation or destruction of an equilibrium.

2.2.2. Period-doubling bifurcation

A period-doubling (flip) bifurcation occurs when one eigenvalue crosses the unit circle through

$$\lambda = -1. \tag{43}$$

Replacing $\lambda = -1$ into Eq. (30) yields

$$1 + T + \Delta = 0. \tag{44}$$

Substituting the analytical expressions for the trace and determinant,

$$1 + A(1 + B) + B = 0, \tag{45}$$

or equivalently,

$$(1 + B)(1 + A) = 0. \tag{46}$$

Thus, excluding the degenerate case $1 + B = 0$, the flip bifurcation occurs whenever

$$F'(x^*, \mu) = -1. \tag{47}$$

Crossing this boundary causes the fixed point to lose stability and a stable orbit of period two is created, initiating the classical period-doubling route toward chaos.

2.2.3. Neimark–Sacker bifurcation

A Neimark–Sacker bifurcation takes place when the two eigenvalues form a complex-conjugate pair located on the unit circle,

$$\lambda_{1,2} = e^{\pm i\theta}, \qquad \theta \neq 0, \pi. \tag{48}$$

Since

$$\lambda_1 \lambda_2 = \Delta, \tag{49}$$

the condition for the eigenvalues to have unit modulus is simply

$$\Delta = 1. \tag{50}$$

Using Eq. (35),

$$1 + a\left(1 - 3by^{*2}\right) = 1, \tag{51}$$

which immediately yields

$$y^* = \pm\frac{1}{\sqrt{3b}}, \tag{52}$$

provided that $b > 0$.

The additional requirement

$$|T| < 2 \tag{53}$$

guarantees that the eigenvalues remain complex during the crossing. At this bifurcation the fixed point loses stability through the birth of a closed invariant curve, which typically gives rise to quasiperiodic dynamics.

The above expressions constitute explicit analytical bifurcation conditions for the Gumowski–Mira map. In particular, Eqs. (42) and (47) demonstrate that the saddle–node and period-doubling bifurcations are completely determined by the derivative of the nonlinear function evaluated at the equilibrium point, whereas the Neimark–Sacker bifurcation depends directly on the equilibrium coordinate through Eq. (52).

3. Parameter space

Before examining the parameter space in detail, it is useful to explore how the dynamical variables change as selected parameters vary. This step helps anticipate whether the parameter space will contain organized structures. Fig. 2 presents the bifurcation diagram of the variable x as a function of the parameter μ, using the initial condition $(x_0, y_0) = (0, 0.5)$ and fixed parameters $a = 0.35$ and $b = 0.1$. Throughout this work, all simulations were performed using the fixed initial condition $(x_0, y_0) = (0, 0.5)$, which is kept unchanged for every parameter combination to ensure a consistent characterization of the global organization of the parameter space. This approach is commonly adopted in parameter-space analyses to provide a systematic comparison of the dynamical regimes across the parameter space. We note, however, that in parameter regions exhibiting multistability, different initial conditions may converge to distinct coexisting attractors. Consequently, the parameter-space diagrams presented here characterize the attractors associated with the chosen initial condition and do not necessarily represent all possible coexisting solutions. A systematic investigation based on multiple or random initial conditions, together with the corresponding basin structures, is beyond the scope of the present work and will be addressed in future studies. For all simulations, an initial transient of 1000 iterations was discarded to eliminate the influence of the initial conditions. The attractor properties were then computed from the subsequent 2000 iterations. Representative parameter values, including cases located close to bifurcation boundaries, were also inspected using longer transients and integration times, yielding no observable changes in the identified asymptotic dynamics. These numerical parameters therefore provide a reliable characterization of the parameter space while maintaining a feasible computational cost for the large-scale simulations performed in this work.

We also compute the Lyapunov exponents, defined according to [51] as

$$\lambda_j = \lim_{n\to\infty} \frac{1}{n} \ln|\Lambda_j|, \qquad j = 1, 2, \tag{54}$$

where Λ_j are the eigenvalues of the matrix $M = \prod_{i=1}^{n} J_i(x_i, y_i)$, and J_i denotes the Jacobian of the map evaluated along the orbit. A positive λ_j indicates chaotic dynamics, while periodic attractors give negative exponents. Values close to zero correspond to bifurcation points.

The bifurcation diagram reveals alternating regions of periodic and chaotic behavior. A notable feature is the presence of a period bubble [52–54], where a periodic branch undergoes a sequence of bifurcations before returning to its original periodicity. Although a detailed mathematical characterization of this structure is beyond the scope of the present work, its occurrence suggests the existence of rich local bifurcation phenomena that deserve further investigation. More generally, the complexity of the bifurcation diagram indicates that the corresponding two-parameter space may also exhibit intricate organizations of periodic and chaotic regions. The supplementary material provides an interactive visualization of the bifurcation diagram for exploring additional parameter values (see Section "Data Availability").

Following these preliminary observations, Fig. 3 displays the structure of the parameter space of the Gumowski–Mira map with $a = 0.35$. To construct this figure, we divide the parameter space $(\mu, b) \in [-0.6, 0.6] \times [0, 1]$ into a uniform grid of 1000×1000 points, producing 10^6 parameter combinations. For each pair (μ, b), we compute the largest Lyapunov exponent λ, shown in Fig. 3(a). The color scale ranges from black to yellow for periodic dynamics (negative exponents) and from blue to green for chaotic dynamics (positive exponents). We also determine the period of the attractor for each parameter pair, shown in Fig. 3(b). The period of the attractor is determined by detecting recurrences of the trajectory after discarding the transient dynamics. Starting from the first point of the post-transient orbit, the subsequent iterates are compared with this reference state using an absolute tolerance of 10^{-10}. A candidate period is identified when the trajectory returns to the reference state within the prescribed tolerance. To avoid spurious

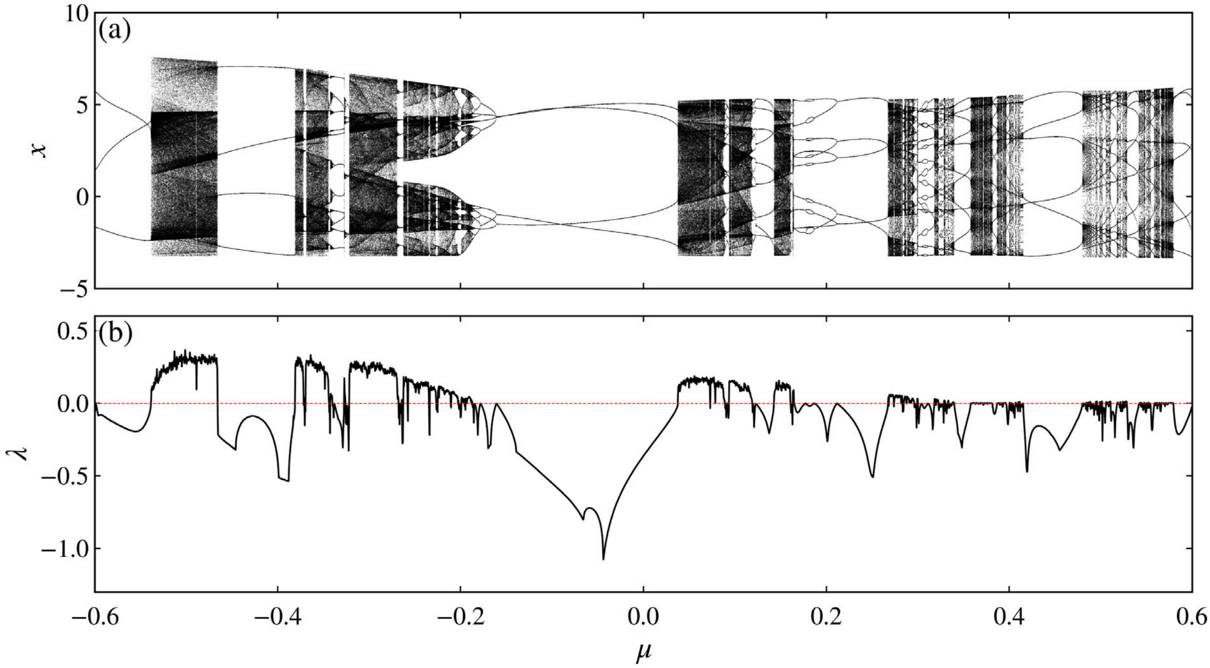

Fig. 2. Bifurcation diagram of the variable x as a function of the parameter μ for fixed values $a = 0.35$ and $b = 0.1$. Periodic windows and chaotic regions alternate as μ varies, indicating the presence of complex transitions in the dynamics.

detections caused by numerical round-off or near recurrences, a period is accepted only after the same recurrence is consistently verified in three consecutive cycles. The search evaluates periods up to a maximum value of $P_{\max} = 1000$. For visualization purposes, periods up to $P = 19$ are assigned distinct colors, whereas all periods $P \geq 20$ are grouped into a single color category. If no valid period is detected within the prescribed search window, the trajectory is treated as non-periodic, a category that includes chaotic dynamics, quasi-periodic motion, and unresolved periodic orbits with periods exceeding the search limit. In the period diagrams, these non-periodic trajectories, together with diverging orbits, are represented by the white regions. The horizontal dashed line shown in Fig. 3(b) corresponds to the parameter slice $b = 0.1$, along which the one-parameter bifurcation diagram presented in Fig. 2 was computed, thereby providing a direct correspondence between the one- and two-parameter analyses. The supplementary material includes a video exploring the parameter space for several values of a, as well as parameter-space visualizations for $\mu \times a$ and $a \times b$ (see Section “Data Availability”).

By varying one parameter while keeping the other fixed or by varying both simultaneously, we observe transitions from chaotic to periodic dynamics and then back to chaotic dynamics, creating a collection of periodic windows. This behavior produces several structures in the parameter space, reflecting the system’s complexity. We focus on three specific regions marked by the boxes labeled A, B, and C in Fig. 3(a). These regions are shown in detail in Fig. 4(a-c), where subindex 1 corresponds to the Lyapunov exponent analysis and subindex 2 to the period analysis. The letter labels match the boxes in Fig. 3.

A well-known organization of the parameter space in such systems is given by shrimp-shaped domains. Such structures were first systematically reported by Gallas [2,3] and correspond to islands of stable periodic dynamics in two-parametric spaces of nonlinear maps. Their characteristic geometry consists of a central stability region surrounded by narrow extensions. The main body of the shrimp is associated with a main periodic region followed by an infinite sequence of period-doubling bifurcations leading to chaos. The observed bifurcations follow the rule $P_0 \times 2^m$, where P_0 denotes the period of the main stability region and m is a non-negative integer. This organization has been reported in several two-dimensional maps and is associated with self-similar replication of stability islands across the parameter space [55–60]. The main periods P_0 of these structures are 11, 14, and 15.

Figs. 4(a_1) and (a_2) display a clear shrimp-shaped structure with the standard morphology reported in the literature. The main periodic body is well defined, followed by its period-doubling cascade toward chaotic dynamics. In the vicinity of this structure, additional stability regions of different sizes and shapes are observed. These secondary structures are related to the self-similarity of the parameter space, appearing as scaled replicas of the primary shrimp. Such a hierarchical organization is a characteristic feature of shrimp structures and has been reported in a wide variety of dissipative nonlinear systems, suggesting that it represents a universal organization of parameter spaces rather than a peculiarity of the Gumowski–Mira map [3,4,8].

In contrast, Figs. 4(b_1) and (b_2) show a stability region that does not exhibit the typical shrimp geometry. Despite this difference, its internal organization follows the same period-doubling rule. The main periodic region has period $P_0 = 9$, and successive bifurcations lead to periods $9 \to 18 \to 36 \to 72 \to 144 \to \cdots$, with the periods obeying the shrimp rule $P_0 \times 2^m$. This observation

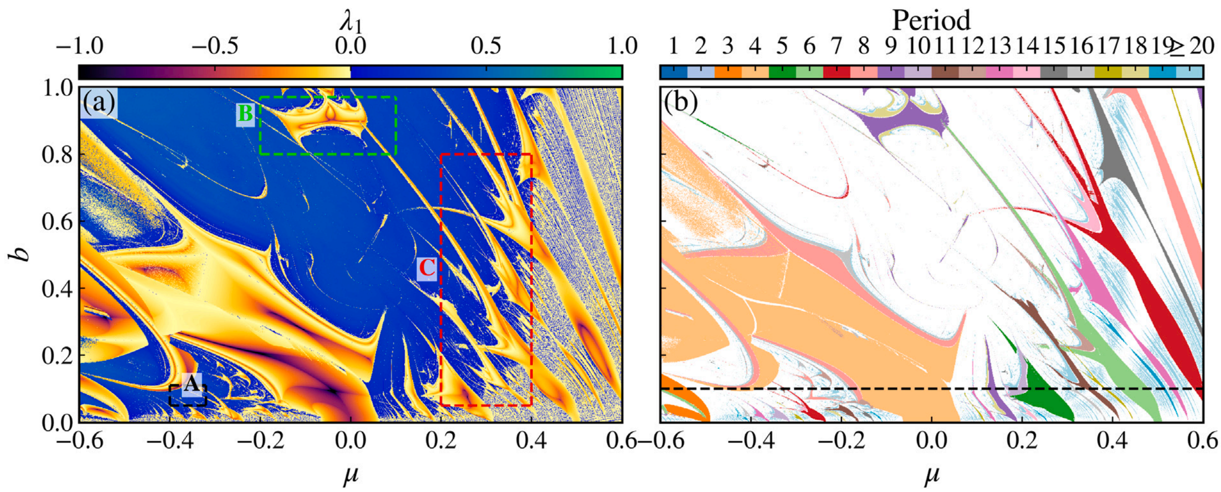

Fig. 3. Parameter space of the Gumowski–Mira map in the (μ, b) plane computed for fixed parameter $a = 0.35$ and initial condition $(x_0, y_0) = (0, 0.5)$ over the range $\mu \in [-0.6, 0.6]$ and $b \in [0, 1]$. Panel (a) shows the largest Lyapunov exponent (λ_1), with a color scale capped between -1 and 1 highlighting periodic regions embedded in chaotic backgrounds. Panel (b) shows the corresponding attractor period. The discrete color scale explicitly resolves periods from 1 up to 19, with periods ≥ 20 grouped into the maximum category, while white denotes non-periodic (chaotic and diverging) orbits. The dashed boxes labeled A, B, and C indicate regions selected for detailed analysis with the following specific coordinates: Region A bounds $\mu \in [-0.4, -0.32]$ and $b \in [0.05, 0.11]$; Region B bounds $\mu \in [-0.2, 0.1]$ and $b \in [0.8, 0.97]$; and Region C bounds $\mu \in [0.2, 0.4]$ and $b \in [0.05, 0.8]$. The horizontal dashed line in panel (b) indicates the parameter slice at $b = 0.1$ used to construct the one-parameter bifurcation diagram shown in Fig. 2.

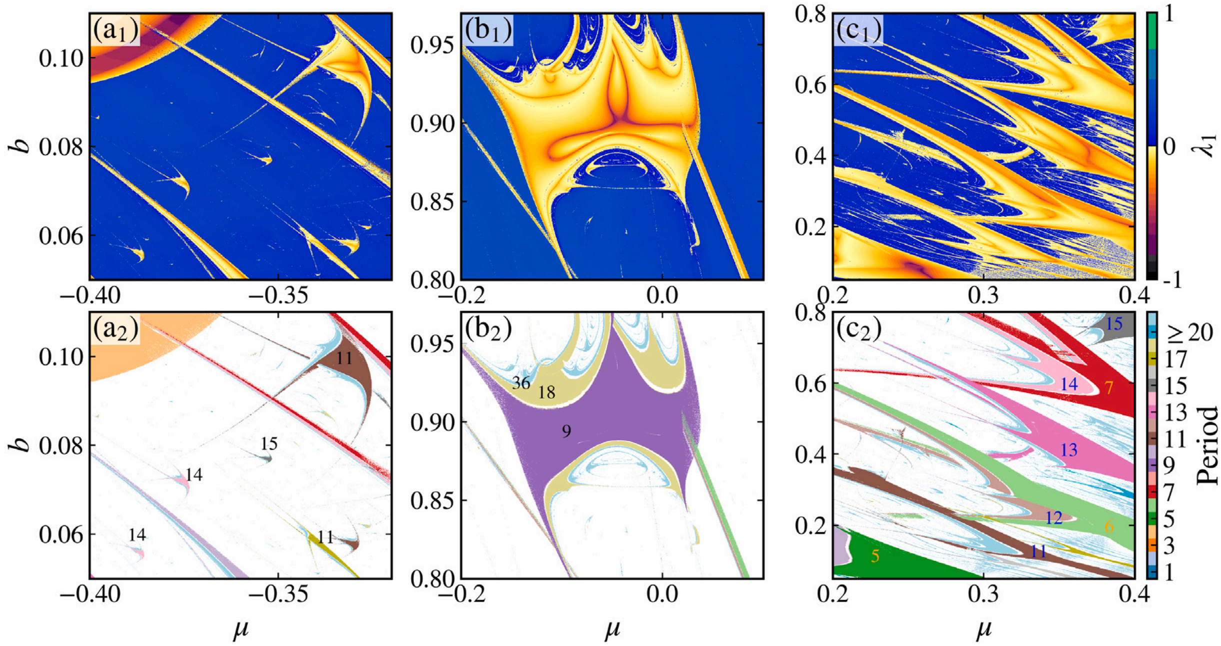

Fig. 4. Magnifications of the regions A, B, and C marked in Fig. 3. Top row (a_1–c_1) show the Lyapunov exponent, while bottom row (a_2–c_2) show the corresponding period of the attractor. Panels (a_1) and (a_2) displays a typical shrimp-shaped structure with a period-doubling cascade and self-similar structures. Panels (b_1) and (b_2) shows a non-standard stability region that still follows a period-doubling sequence. Panels (c_1) and (c_2) reveals tongue-like periodic domains organized along paths of increasing period.

suggests that the universal bifurcation hierarchy associated with shrimp structures is preserved even in stability regions with non-standard geometries.

A different organization of the parameter space is observed in Figs. 4(c_1) and (c_2), where multiple tongue-like periodic domains regions dominate the dynamics. Along these tongues, one can identify paths of increasing periodicity. Two distinct patterns are observed and indicated by orange and blue paths. Along the orange path, the period increases from 5 to 6 and then to 7, after which the subsequent tongue is not observed before the sequence resumes at higher periods. One possible explanation is that the intermediate periodic tongues become extremely narrow, preventing them from being resolved at the numerical resolution employed in the present computations. Such an interpretation is consistent with previous studies showing that higher-order Arnold tongues may become progressively narrower and compete with neighboring tongues in parameter space [61]. Nevertheless, a detailed investigation of the mechanism responsible for this apparent tongue-skipping behavior remains an interesting topic for future work. Along the blue path, the system passes consecutively through all tongues, with the orbit transitioning from the base of one tongue to the head of the next. Along this path, the period increases monotonically following the sequence 11, 12, 13, 14, 15, and so on. These patterns indicate that multiple mechanisms of period organization coexist in the parameter space of the Gumowski–Mira map, not just shrimp structures.

4. Conclusion

In this work, we investigated the two-parameter space of the Gumowski–Mira map using Lyapunov exponent and period analyses to characterize the organization of its dynamical regimes. The results reveal a rich and intricate structure composed of periodic, quasi-periodic, and chaotic regions, organized in a highly nontrivial manner.

The parameter space exhibits well-defined shrimp-shaped stability domains embedded in chaotic regions. These shrimps display the standard organization reported in other nonlinear maps, with a main periodic region followed by infinite sequences of period-doubling bifurcations obeying the rule $P_0 \times 2^m$. In addition to classical shrimp geometries, we identified stability regions that do not present the typical shrimp shape but still follow the universal Feigenbaum period-doubling route to chaos, indicating that the bifurcation mechanism is preserved even when the geometric outline changes.

Furthermore, we observed regions dominated by tongues, where the dynamics is organized according to period-adding sequences. Distinct paths of increasing periodicity were identified, including both linear progressions through consecutive tongues and paths that skip intermediate tongues. These observations demonstrate that multiple mechanisms of period organization coexist in the parameter space of the Gumowski–Mira map.

Overall, our results demonstrate that the parameter space of the Gumowski–Mira map exhibits several characteristic features that are commonly observed in dissipative nonlinear systems, including self-similar shrimp structures, hierarchical parameter-space organization, and period-doubling cascades consistent with the classical Feigenbaum scenario. The recurrence of these features across a wide variety of nonlinear models suggests that they constitute robust organizing principles of dissipative dynamics rather than properties unique to the Gumowski–Mira map [62].

CRediT authorship contribution statement

Daniel Borin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Diego F.M. Oliveira:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was financed, in part, by the São Paulo Research Foundation (FAPESP), Brazil, Process Numbers #2024/06749-8 and #2022/03612-6.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. Additionally, the supplementary material is accessible via the interactive platform at https://gumowskimira.streamlit.app.

References

[1] G.C. Layek, An Introduction to Dynamical Systems and Chaos, second ed., Springer, 2024.
[2] J.A.C. Gallas, Structure of the parameter space of the Hénon map, Phys. Rev. Lett. 70 (1993) 2714–2717.
[3] J.A. Gallas, Dissecting shrimps: results for some one-dimensional physical models, Phys. A 202 (1) (1994) 196–223.
[4] D.F.M. Oliveira, E.D. Leonel, Parameter space for a dissipative Fermi–Ulam model, New J. Phys. 13 (12) (2011) 123012.
[5] D.F.M. Oliveira, M. Robnik, E.D. Leonel, Statistical properties of a dissipative kicked system: Critical exponents and scaling invariance, Phys. Lett. A 376 (5) (2012) 723–728.
[6] D.F.M. Oliveira, M. Robnik, E.D. Leonel, Scaling invariance in a dissipative standard map, Phys. Lett. A 375 (38) (2011) 3365–3369.
[7] M. Hansen, D.R. da Costa, D.F. Oliveira, E.D. Leonel, Statistical properties for a dissipative model of relativistic particles in a wave packet: A parameter space investigation, Appl. Math. Comput. 238 (2014) 387–392.
[8] R. Stoop, S. Martignoli, P. Benner, R.L. Stoop, Y. Uwate, Shrimps: Occurrence, scaling and relevance, Int. J. Bifurc. Chaos 22 (10) (2012) 1230032.
[9] L. Glass, R. Perez, Fine structure of phase locking, Phys. Rev. Lett. 48 (1982) 1772–1775.
[10] R. Perez, L. Glass, Bistability, period doubling bifurcations and chaos in a periodically forced oscillator, Phys. Lett. A 90 (9) (1982) 441–443.
[11] P. Gaspard, R. Kapral, G. Nicolis, Bifurcation phenomena near homoclinic systems: A two-parameter analysis, J. Stat. Phys. 35 (5) (1984) 697–727.
[12] S. Fraser, R. Kapral, Analysis of flow hysteresis by a one-dimensional map, Phys. Rev. A 25 (1982) 3223–3233.
[13] M. Markus, B. Hess, Lyapunov exponents of the logistic map with periodic forcing, Comput. Graph. (1989).
[14] J.P. Carcasses, C. Mira, M. Bosch, C. Simó, J.C. Tatjer, Crossroad area-spring area transition (I) parameter plane representation, Int. J. Bifurc. Chaos 01 (01) (1991) 183–196.
[15] E.N. Lorenz, Compound windows of the Hénon-map, Phys. D: Nonlinear Phenom. 237 (13) (2008) 1689–1704.
[16] N.C. Pati, P. Datta, B. Ghosh, Exploring quasi-periodic shrimps in the parameter space of a discrete-time food chain model, Chaos: An Interdiscip. J. Nonlinear Sci. 35 (2) (2025) 023101.
[17] N.C. Pati, Spiral organization of quasi-periodic shrimp-shaped domains in a discrete predator–prey system, Chaos: An Interdiscip. J. Nonlinear Sci. 34 (8) (2024) 083126.
[18] V.I. Arnol'd, Small denominators. I. Mapping the circle onto itself, Izv Ross Akad Nauk. Seriya Mat 25 (1) (1961) 21.
[19] R. Simile Baroni, R. Egydio de Carvalho, C.E.P. Abreu, R.O. Medrano-T, Global dynamics and asymmetric fractal dimension in a nontwist circle map, Chaos: An Interdiscip. J. Nonlinear Sci. 35 (2) (2025) 023122.
[20] G.C. Layek, N.C. Pati, Organized structures of two bidirectionally coupled logistic maps, Chaos 29 (9) (2019) 093104.
[21] V.E. Camargo, A.S. Amaral, A.F. Crepaldi, F.F. Ferreira, Synchronization and bifurcation in an economic model, Chaos: An Interdiscip. J. Nonlinear Sci. 32 (10) (2022) 103112.
[22] N.C. Pati, S. Garai, M. Hossain, G.C. Layek, N. Pal, Fear induced multistability in a predator-prey model, Int. J. Bifurc. Chaos 31 (10) (2021) 2150150.
[23] M. Hossain, R. Kumbhakar, N. Pal, J. Kurths, Structure of parameter space of a three-species food chain model with immigration and emigration, Nonlinear Dynam. 111 (15) (2023) 14565–14582.
[24] M.R. Gallas, M.R. Gallas, J.A. Gallas, Distribution of chaos and periodic spikes in a three-cell population model of cancer, Eur. Phys. J. Spec. Top. 223 (11) (2014) 2131–2144.
[25] C.d.S. Rodrigues, C.G.P. dos Santos, R.C.C. de Miranda, E. Parma, H. Varela, R. Nagao, A numerical investigation of the effect of external resistance and applied potential on the distribution of periodicity and chaos in the anodic dissolution of nickel, Phys. Chem. Chem. Phys. 22 (2020) 21823–21834.
[26] C.A.T. Chávez, S. Curilef, Discontinuous spirals of stability in an optically injected semiconductor laser, Chaos: An Interdiscip. J. Nonlinear Sci. 30 (5) (2020) 053107.
[27] C. Bonatto, J.C. Garreau, J.A.C. Gallas, Self-similarities in the frequency-amplitude space of a loss-modulated CO_2 laser, Phys. Rev. Lett. 95 (2005) 143905.
[28] S. Garai, M. Hossain, S. Karmakar, N. Pal, Chaos, periodic structures, and multistability: Complex dynamical behaviors of an eco-epidemiological model in parameter planes, Chaos: An Interdiscip. J. Nonlinear Sci. 33 (8) (2023) 083115.
[29] B.-K. Hou, J.-S. Gao, X.-B. Rao, S.-L. Ding, Novel organizational patterns of stability phases in a single-species population model: Chiral tree, spikes adding-doubling complexification cascade, Nonlinear Dynam. 112 (19) (2024) 17611–17626.
[30] J.C. Xavier, P.C. Rech, Regular and chaotic dynamics of the Lorenz-Stenflo system, Int. J. Bifurc. Chaos 20 (01) (2010) 145–152.
[31] M.N. Mahmud, Z. Siri, J.A. Vélez, L.M. Pérez, D. Laroze, Chaotic convection in an oldroyd viscoelastic fluid in saturated porous medium with feedback control, Chaos: An Interdiscip. J. Nonlinear Sci. 30 (7) (2020) 073109.
[32] A.C. Mathias, P.C. Rech, Hopfield neural network: The hyperbolic tangent and the piecewise-linear activation functions, Neural Netw. 34 (2012) 42–45.
[33] P.C. Rech, Period-adding and spiral organization of the periodicity in a Hopfield neural network, Int. J. Mach. Learn. Cybern. 6 (1) (2015) 1–6.
[34] B. Raphaldini, E.S. Medeiros, D. Ciro, D.R. Franco, R.I.F. Trindade, Geomagnetic reversals at the edge of regularity, Phys. Rev. Res. 3 (2021) 013158.
[35] R. Varga, K. Klapcsik, F. Hegedűs, Route to shrimps: Dissipation driven formation of shrimp-shaped domains, Chaos Solitons Fractals 130 (2020) 109424.
[36] R. Stoop, P. Benner, Y. Uwate, Real-world existence and origins of the spiral organization of shrimp-shaped domains, Phys. Rev. Lett. 105 (2010) 074102.
[37] E.R. Viana Jr., R.M. Rubinger, H.A. Albuquerque, A.G. de Oliveira, G.M. Ribeiro, High-resolution parameter space of an experimental chaotic circuit, Chaos: An Interdiscip. J. Nonlinear Sci. 20 (2) (2010) 023110.
[38] F.F.G. de Sousa, R.M. Rubinger, J.C. Sartorelli, H.A. Albuquerque, M.S. Baptista, Parameter space of experimental chaotic circuits with high-precision control parameters, Chaos: An Interdiscip. J. Nonlinear Sci. 26 (8) (2016) 083107.
[39] F. Prebianca, D.W. Marcondes, H.A. Albuquerque, M.W. Beims, Exploring an experimental analog Chua's circuit, Eur. Phys. J. B 92 (6) (2019) 134.
[40] W. Cui, L. Zhao, Y. Ge, K. Xu, A generalized van der Pol nonlinear model of vortex-induced vibrations of bridge decks with multistability, Nonlinear Dynam. 112 (1) (2024) 259–272.
[41] X. Liu, L. Zhao, J. Jin, A noise-tolerant fuzzy-type zeroing neural network for robust synchronization of chaotic systems, Concurr. Comput.: Pr. Exp. 36 (22) (2024) e8218.
[42] J. Jin, W. Chen, A. Ouyang, F. Yu, H. Liu, A time-varying fuzzy parameter zeroing neural network for the synchronization of chaotic systems, IEEE Trans. Emerg. Top. Comput. Intell. 8 (1) (2024) 364–376.
[43] C. Mira, I. Gumowski and a toulouse research group in the ''Prehistoric'' times of chaotic dynamics, in: R.H. Abraham, Y. Ueda (Eds.), The Chaos Avant-Garde: Memories of the Early Days of Chaos Theory, World Scientific, Singapore, 2000, pp. 95–197.
[44] G. Hanchinamani, L. Kulkarni, An efficient image encryption scheme based on quintuple encryption using Gumowski–Mira and tent maps, Int. J. Contents 11 (4) (2015) 56–69.
[45] X. Yan, J. Liu, J. Zhang, P. Li, An improved Gumowski–Mira map with symmetric Lyapunov exponents, Int. J. Bifurc. Chaos 32 (14) (2022) 2250099.
[46] I. Gumowski, C. Mira, Recurrences and discrete dynamical systems, in: Lecture Notes in Mathematics, vol. 809, Springer, Berlin, Heidelberg, 1980.
[47] L.M. Saha, M.K. Das, M. Budhraja, Characterization of attractors in Gumowski–Mira map using fast Lyapunov indicators, Pramana 67 (1) (2006) 151–161.
[48] S.N. Prasad, K.R. Meena, A.A. Ansari, Complexity dynamics of Gumowski–Mira map, Appl. Appl. Math. 15 (1) (2020) 273–281.
[49] C. Mira, Chaotic Dynamics: From the One-Dimensional Endomorphism to the Two-Dimensional Diffeomorphism, World Scientific, Singapore, 1987.
[50] C. Mira, L. Gardini, A. Barugola, J.-C. Cathala, Chaotic dynamics in two-dimensional noninvertible maps, in: World Scientific Series on Nonlinear Science, Series A, vol. 20, World Scientific, Singapore, 1996, p. 632.
[51] J.-P. Eckmann, D. Ruelle, Ergodic theory of chaos and strange attractors, Rev. Modern Phys. 57 (3) (1985) 617.

[52] D. Borin, M. Rolim Sales, E.D. Leonel, D.F.M. de Oliveira, On the dynamics of the q-Tsallis Gauss iterated map, Phys. A 686 (2026) 131352.

[53] E. Liz, A. Ruiz-Herrera, The hydra effect, bubbles, and chaos in a simple discrete population model with constant effort harvesting, J. Math. Biol. 65 (5) (2012) 997–1016.

[54] M. Bier, T.C. Bountis, Remerging feigenbaum trees in dynamical systems, Phys. Lett. A 104 (5) (1984) 239–244.

[55] D. F. M. Oliveira, Mapping chaos: Bifurcation patterns and shrimp structures in the Ikeda map, Chaos: An Interdiscip. J. Nonlinear Sci. 34 (12) (2024) 123137.

[56] M. Rolim Sales, M. Mugnaine, E.D. Leonel, I.L. Caldas, J.D. Szezech Jr., Shrinking shrimp-shaped domains and multistability in the dissipative asymmetric kicked rotor map, Chaos: An Interdiscip. J. Nonlinear Sci. 34 (11) (2024) 113129.

[57] Rajni, B. Ghosh, Arnold tongues, shrimp structures, multistability, and ecological paradoxes in a discrete-time predator–prey system, Chaos: An Interdiscip. J. Nonlinear Sci. 34 (12) (2024) 123103.

[58] R.B. do Carmo, J.R. Rios Leite, F.M. de Aguiar, One-dimensional invariant measure in periodicity hubs: A tribute to professor Jason A. C. Gallas, Chaos: An Interdiscip. J. Nonlinear Sci. 35 (2) (2025) 023116.

[59] R. Kumbhakar, S. Karmakar, N. Pal, J. Kurths, Shrimp-shaped structure and period-bubbling route to chaos in a one-dimensional economic model, Chaos: An Interdiscip. J. Nonlinear Sci. 34 (10) (2024) 103115.

[60] S.L.T. de Souza, A.M. Batista, R.O. Medrano-T, I.L. Caldas, Quasiperiodic shrimp-shaped domains in intrinsically coupled oscillators, Chaos: An Interdiscip. J. Nonlinear Sci. 34 (12) (2024) 123146.

[61] D. García-Álvarez, A. Bahraminasab, A. Stefanovska, P.V.E. McClintock, High-order synchronization, transitions, and competition among arnold tongues in a rotator under harmonic forcing, Phys. Rev. E 77 (2008) 056203.

[62] W.M. Façanha, B. Oldeman, L. Glass, Bifurcation structures in two-dimensional maps: The endoskeletons of shrimps, Phys. Lett. A 377 (18–19) (2013) 1264–1268.